



**PROFESSIONAL TRADING
PLATFORM**

COMPREHENSIVE USER MANUAL

**Advanced Multi-Layered Trading
System**

X13.05

80+ Pages of Professional Training

& Complete Reference Guide

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MASTER THE PLATFORM

Complete step-by-step guidance from
installation to advanced trading strategies



COMPREHENSIVE USER MANUAL

Advanced Multi-Layered Trading Platform

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1. GETTING STARTED

Welcome to the Platform

The Advanced Multi-Layered Trading Platform is a professional-grade system combining three complementary trading strategies with machine learning and comprehensive risk management.

Key Concepts:

Strategy: A predefined set of rules for entering and exiting trades based on technical analysis

Trading Mode: The environment where you trade (Demo with virtual money, Live with real money, or Backtest with historical data)

Indicator: A technical tool (EMA, RSI, ADX, etc.) that helps identify trading opportunities

Risk Management: Rules to protect your capital (stop losses, position sizing, profit targets)

ML Model: Artificial intelligence algorithm that learns to predict price movements

System Requirements

Minimum:

- Windows 10/11 or Linux

- Python 3.8+
- 4GB RAM
- Stable internet connection

Recommended:

- Windows 10/11 64-bit
 - Python 3.10+
 - 8GB+ RAM
 - Fast internet (for live trading)
-

Installation Steps

Step 1: Install Python

- Download Python 3.8+ from python.org
- Make sure to check “Add Python to PATH” during installation

Step 2: Install Required Libraries

Open Command Prompt/Terminal and run:

```
pip install pandas numpy talib ccxt backtesting
scikit-learn xgboost tensorflow pandas
_ta openpyxl
```

Step 3: Launch the Platform

python [main_New_MACD_HybridScoreapp](#)_New.py

The main window will open with the trading interface.

2. INSTALLATION & SETUP

API Configuration

The platform connects to cryptocurrency exchanges via APIs. You need API credentials for live trading.

For OKX Exchange (Recommended):

1. **Create OKX Account:**
 - Go to okx.com
 - Sign up and verify identity
 - Enable 2FA (two-factor authentication)
2. **Generate API Key:**
 - Go to Account → API Management
 - Create new key
 - Note: API Key, Secret Key, Passphrase
 - Set appropriate permissions (trading, read-only recommended for safety)
3. **Configure in Platform:**
 - Create config.json file in app directory:

```
{
  "live": {
    "api_key": "your_okx_api_key",
    "api_secret_key": "your_okx_secret",
    "passphrase": "your_okx_passphrase"
  },
  "demo": {
    "api_key": "demo_key_not_needed",
    "api_secret_key": "demo_secret_not_need
ed",
```

```
    "passphrase": "demo_pass_not_needed"
  },
  "backtest": {
    "api_key": "backtest_key_not_needed",
    "api_secret_key": "backtest_secret_not_needed",
    "passphrase": "backtest_pass_not_needed"
  }
}
```

4. Test Connection:

- Click “Check Connection” button in the platform
- You’ll see “Connection successful!” if configured correctly

Security Best Practices:

⚠ CRITICAL SECURITY NOTES: - Never share your API keys - Use API key restriction features (IP whitelist) - Create a separate API key just for trading (limited permissions) - Store config.json securely (not in version control) - Regularly rotate your API keys - Enable 2FA on exchange account - Start with small amounts in live mode

3. CORE FEATURES OVERVIEW

Main Components

A. Trading Dashboard

Located on the left side of the screen, displays: - Current trading status (BUY/SELL/PARKING) - Market symbols and timeframes - Strategy selection - ML model options - Volume strength gauge - Trading settings - Trading statistics

B. Price Chart

Located on the right side: - Real-time candlestick chart - Technical indicators overlaid - Price entry/exit markers - Support/resistance levels

C. Trading Controls

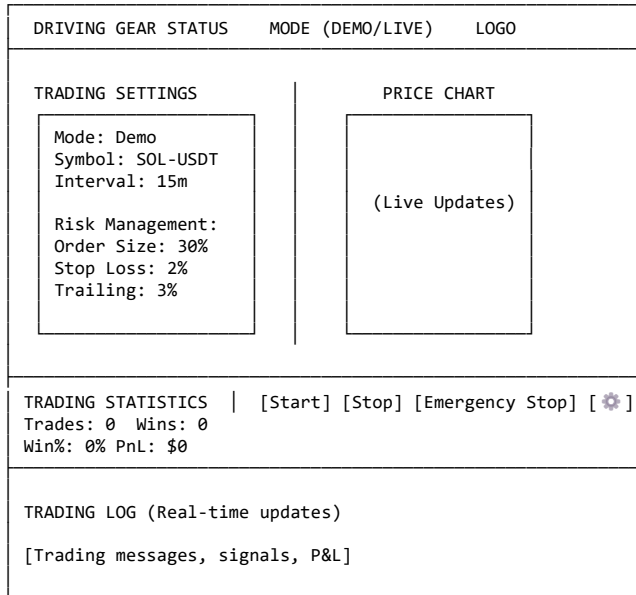
- Start/Stop buttons
- Emergency Stop button
- Check Connection button
- Settings button
- Mode selector (Live/Demo/Backtest)

D. Trading Log

Bottom section shows: - Real-time trading messages - Entry/exit signals - P&L updates - System notifications - Status messages

4. USER INTERFACE GUIDE

Main Window Layout



Control Descriptions

Mode Selector

- **Demo:** Trade with \$1,000 virtual money (learning)
- **Live:** Real trading with actual capital (requires API config)
- **Backtest:** Test with historical data (validation & optimization)

Symbol Selector

- **SOL-USDT:** Solana (default)
- **BTC-USDT:** Bitcoin

- **ETH-USDT:** Ethereum
- Custom symbols supported

Interval Selector

- **1m, 5m, 15m, 30m:** Short-term trading
- **1H, 4H:** Medium-term
- **1D:** Long-term

*Shorter intervals = more trades, higher risk
Longer intervals = fewer trades, lower noise*

Strategy Selector

- **Momentum:** Best for uptrends (90-130% annual potential)
- **Kalman:** Best for mean-reversion (80-120% annual potential)
- **Enhanced:** Hybrid (flexible, custom parameters)

ML Integration

- **Enable ML:** Turn machine learning on/off
 - **Model Selection:** Choose Random Forest, XGBoost, or LSTM
 - **Confidence Threshold:** Set 55-95% (higher = fewer, higher-conviction trades)
 - **Auto-Execute:** Automatically trade on ML signals
-

5. THREE TRADING STRATEGIES

Strategy 1: MOMENTUM ~~v7.5~~v5.2

Overview

Designed for trending markets with strong momentum. Uses multiple confirmation filters to enter only high-quality trades.

Key Principles:

1. **Trend Identification:** Price > EMA_Fast > EMA_Mid > EMA_Slow
2. **Trend Strength:** ADX must be ≥ 25 (hard minimum, ADX 25–34 optimal ~~>18~~ (preferably >25))
3. **Entry Timing:** When multiple confirmation signals align
4. **Exit Strategy:** Progressive profit taking + trailing stops

Entry Conditions (Must ALL Be True):

✓ Price above 3 EMAs in correct order (Close > EMA9 > EMA21 > EMA50) ✓ ADX ≥ 25 (hard minimum — avoids choppy ATR-wide stops ~~>18~~ (trend exists)) ✓ RSI between 45–58 (pullback zone, 40–70 ~~(not extreme)~~) ✓ Volume $\geq 1.2 \times 3 \times$ average ✓ MACD line above signal line ✓ Price within 1.5% of EMA9 (not over-extended) ✓ Daily trend filter: Close above 50-day EMA ~~Kalman~~ strength > 0.05 ✓ Market NOT ranging (8-check filter passes)

Entry Quality Scoring:

- Score $\geq 88/100$ (Tier 2): Exceptional setup
— 1.5% risk, max position size
- Score 75-87/100 (Tier 1): Good setup —
1.2% risk, standard position size
- Score $< 75/100$ (Rejected): Quality too low
— no trade, near-miss logged for learning
- ~~5/5 signals: Essential filters only = Good entry~~
- ~~6/7 signals: +1 Additional confirmation = Better entry~~
- ~~7/7 signals: +2 Additional confirmations = Best entry~~

Risk Management:

- Stop Loss: ~~3.0x~~2.2x ATR below entry
- Trailing Stop: ~~43.0x~~ ATR
- Profit Targets: ~~2R (25%), 4R (25%), 6R (25%), 8R (25%)~~ — 1.5R, 2.5R, 4.0R (scale out)
- Position Size: ~~1.2% risk/5% of account per trade~~ (Tier 1), 1.5% risk/trade (Tier 2)
- Max Hold: 120 bars

Exit Signals (Any One Triggers Exit):

1. **Stop Loss Hit:** Protect capital immediately
2. **Profit Target:** Scale out at predetermined levels
3. **Trailing Stop:** Lock in profits as price rises
4. **Trend Reversal:** EMAs cross down (exit if already profitable)
5. **Overbought:** RSI > 70 and MACD bearish

6. **Time Exit:** Held for 120 bars (too long in one trade)

Performance Expectations:

- Win Rate: 52-60%
- Average Trade: 1.2-1.8% gain
- Sharpe Ratio: 1.3-2.0
- Max Drawdown: 15-20%
- Monthly Trades (Bull): 10-15
- Monthly Trades (Range): 0-2

Best Used:

✅ Bullish markets ✅ 5m-1H timeframes (active trading) ✅ When you can monitor regularly ✅ Strong uptrend environments

Strategy 2: KALMAN TREND

Overview

Uses Kalman filtering to detect trend strength and enters on pullbacks rather than peaks. Excellent for mean-reversion in strong trends.

Key Principles:

1. **Kalman Detection:** Identifies trend strength using Kalman filter
2. **Pullback Entry:** Buys after small retracements (not at tops)
3. **Confirmation:** Entry on pullback back to moving average

4. **Wide Stops:** 1.5% stop loss allows for volatility

Entry Conditions:

✓ Kalman Color = Green (uptrend detected) ✓
Kalman Strength > 70 (strong trend) ✓ EMA_Fast
> EMA_Slow (uptrend) ✓ Price pulling back to
EMA_Fast (pullback detected) ✓ RSI between 40-
65 (not extreme) ✓ Volume > 1.3x average ✓
ADX > 25 (strong trend) ✓ Market NOT ranging

Why Pullbacks?

Traditional momentum buys at peaks (risky).
Kalman strategy waits for small retracement, then
buys on bounce back up. This: - Avoids buying at
peaks - Gets better entry price - Reduces stop loss
distance - Increases win rate

Risk Management:

- Stop Loss: 1.5% below entry (wider than Momentum)
- Trailing Stop: 1.0% for profit protection
- Position Size: 1.5% of account
- Cooldown: 10 bars minimum between trades
- Max Hold: 100 bars

Exit Signals:

1. **Stop Loss:** Hard stop at 1.5% loss
2. **Trailing Stop:** If price retraces 1% from high

3. **Trend Break:** Kalman color turns red
4. **Weak Trend:** ADX drops below 20
5. **Time Based:** 100 bars held
6. **Momentum Loss:** MACD crosses down

Performance Expectations:

- Win Rate: 52-58%
- Average Trade: 1.0-1.5% gain
- Sharpe Ratio: 1.2-1.8
- Max Drawdown: 12-16%
- Monthly Trades: 8-12
- Profitable in: 70% of market conditions

Best Used:

✓ Pullback trading ✓ Strong uptrends ✓ Less active monitoring ✓ 15m-4H timeframes ✓
Steady, consistent trading

Strategy 3: ENHANCED

Overview

Customizable hybrid combining best elements of Momentum and Kalman strategies.

Features:

✓ Fully customizable parameters ✓ Combine Momentum + Kalman logic ✓ Market regime detection ✓ Advanced exit conditions ✓ Optional ML confirmation

Use When:

- Testing custom parameters
 - Developing your own strategy
 - Combining multiple indicators
 - Fine-tuning for specific markets
-

6. TRADING MODES

Mode 1: DEMO TRADING

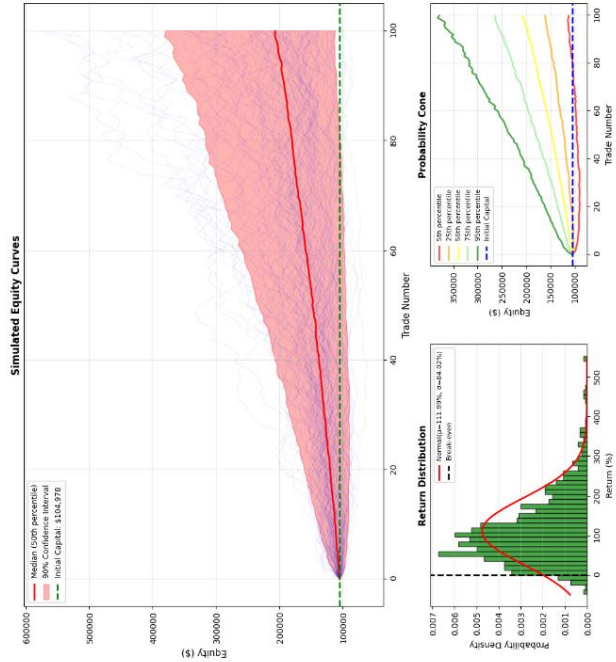
Purpose

Learn the platform without risking real money.

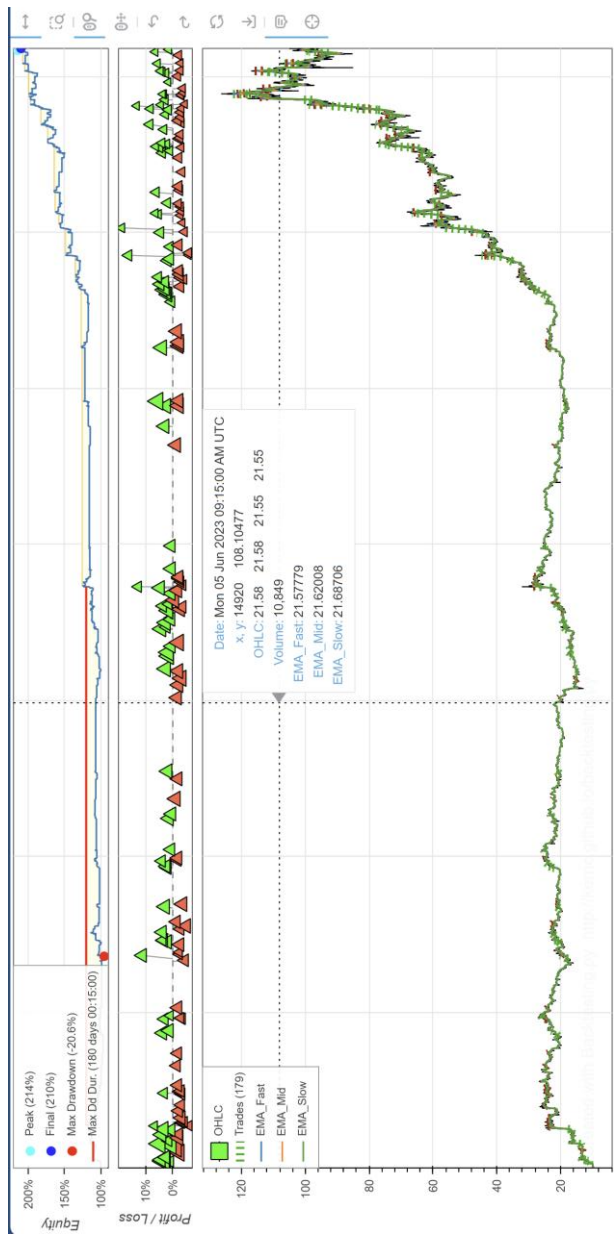
Features:

- Virtual \$1,000 starting capital
- All features available
- Real market data
- Realistic order execution simulation
- Perfect for practice

Monte Carlo Simulation Results



MONTE CARLO SUMMARY	
INITIAL CAPITAL	\$104,976
FINAL CAPITAL	\$272,547
Median	\$206,445
Mean	\$272,547
Profit Prob	97.4%
RETURNS	
Mean	11.19%
StdDev	44.02%
5th-95th	-261.23% - 261.23%
DRAWDOWN	
Mean	21.42%
Median	20.18%
Worst	57.28%
RISK METRICS	
Sharpe Ratio	0.301



Getting Started:

1. Select Mode: **Demo**
2. Choose Symbol: **SOL-USDT** (or preferred)
3. Select Interval: **15m** (recommended for learning)
4. Select Strategy: **Momentum** (start with this)
5. Click **Check Connection** (should show green checkmark)
6. Click **Start Trading**

Demo Mode Benefits:

✓ No financial risk ✓ Full feature access ✓
Learn execution ✓ Test strategies ✓ Build confidence

Demo Recommendations:

- Trade for at least 1 week
 - Complete at least 20-30 trades
 - Understand your strategy deeply
 - Paper trade until you're comfortable
-

Mode 2: LIVE TRADING

Purpose

Trade real cryptocurrency with actual capital.

Prerequisites:

✓ OKX account set up ✓ API keys generated & configured ✓ config.json file created ✓ At least

1-2 weeks demo trading completed 

Understanding of risks

Getting Started:

1. Select Mode: **Live**
2. Enter symbol, interval, strategy (as in Demo)
3. Click **Check Connection** (verify API working)
4. Set Risk Settings:
 - Order Size: 30% (start conservative)
 - Stop Loss: 2%
 - Trailing: 3%
5. Click **Start Trading**

CRITICAL LIVE TRADING RULES:

 **NEVER:** - Trade more than 1% per trade initially - Ignore your stop losses - Trade with money you can't afford to lose - Change parameters mid-trade - Trade without understanding the signal

 **ALWAYS:** - Start with small position sizes - Use hard stop losses - Monitor your first 5-10 trades - Keep a trading journal - Review results daily

Position Sizing for Live:

- Conservative: 0.5-1% risk per trade
- Moderate: 1-1.5% risk per trade
- Aggressive: 1.5-2% risk per trade (maximum)

"If you're not sure, size smaller"

Mode 3: BACKTEST

Purpose

Test strategies using historical data to validate profitability before risking real capital.

How It Works:

1. Select Mode: **Backtest**
2. Choose Symbol: SOL-USDT (default)
3. Set Date Range: From/To dates
4. Select Strategy: Momentum, Kalman, or Enhanced
5. Configure Backtest:
 - **Data Source:** Fetch API Data (or Generated)
 - **Cache:** Use CSV (for speed)
 - **Type:** Standard Backtest or Optimization
 - **Monte Carlo:** Optional (adds probability analysis)
6. Click  **Run Backtest**

Standard Backtest Results:

Final Equity: \$75,432.50

Total Trades: 45

Win Rate: 55.6%

Return: 50.6%

Sharpe Ratio: 1.45

Max Drawdown: 12.3%

Best Trade: +4.5%

Worst Trade: -2.1%

Avg Trade: +1.1%
Profit Factor: 2.34

Interpreting Results:

Good Backtest Signs: - Win Rate: > 50% (better: > 55%) - Sharpe > 1.0 (better: > 1.3) - Max DD < 20% (better: < 15%) - Profit Factor > 2.0 - Consistent monthly returns

Red Flags: - Win rate < 45% - Sharpe < 0.8 - Max drawdown > 30% - Only a few trades (need at least 20) - Profits from only 1-2 trades

Excel Export:

After backtest, three sheets are generated:

Sheet 1: Market Data - OHLCV prices - 30+ indicators - All calculation details - Useful for analysis

Sheet 2: Trades - Entry/exit times & prices - P&L per trade - Win/loss classification - Exit reason - Duration held

Sheet 3: Summary - Performance metrics - Risk statistics - Trade statistics - Analysis tools

7. STRATEGY SELECTION & CONFIGURATION

Selecting Your First Strategy

For Beginners: START WITH MOMENTUM

- Most straightforward logic
- Best documentation
- Good performance
- Easier to understand signals

For Advanced: TRY KALMAN

- More sophisticated entry logic
- Better in mean-reversion environments
- Requires more market understanding
- Potentially more consistent

For Customization: USE ENHANCED

- Build your own rules
 - Combine multiple indicators
 - Advanced users only
-

Parameter Management

The platform has TWO parameter modes:

1. DEFAULT PARAMETERS

- Built-in, proven settings
- Optimized by the developers
- Recommended for most traders
- Can't be modified on-the-fly

2. CUSTOM PARAMETERS

- User-defined settings
- Allows parameter optimization
- More flexibility
- Requires understanding of each parameter

Switching Between Modes:

In Settings Panel → Parameter Tab: - Toggle:

Default Parameters ↔ **Custom Parameters** - If using Custom, click  **Save Custom Parameters** -

To revert: Click  **Reset to Defaults**

Important Momentum Parameters

EMA Configuration:

- ema_fast_period: ~~9~~5 (default) — Responsive to recent price
- ema_mid_period: ~~21~~26 (default) — Medium-term trend
- ema_slow_period: ~~50~~60 (default) — Long-term trend
- **Guideline:** Fast < Mid < Slow always!

Entry Filters:

- adx_min: ~~25~~18 (default) — Hard minimum trend strength (v7.5: raised from 18)
- rsi_entry_min: ~~45~~40 (default) — Pullback zone entry minimum (v7.5: raised from 40) Don't buy oversold

- rsi_entry_max: ~~5870~~ (default) — Pullback zone entry maximum (v7.5: lowered from 70) Don't buy overbought
- volume_min_ratio: 1.3 (default) — Volume confirmation
- kalman_min_strength: ~~0.005~~ (default) — Trend quality proxy (EMA spread-based, not Kalman filter in v7.5)

Risk Management:

- stop_loss_atr_mult: ~~3.0x~~~~2.2x~~ ATR (default) — Stop distance (v7.5: widened from 2.2x)
- trailing_stop_atr_mult: ~~43.0x~~ ATR (default) — Trailing distance (v7.5: widened from 3.0x)
- trailing_stop_pct: 0.04 (4%, default) — Percentage-based
- risk_full_position: ~~0.012 / 0.015~~ (Tier 1/Tier 2.5%, default) — Per-trade risk (v7.5: Tier 1 lowered from 1.5%)
- max_holdBars: 120 (default) — Maximum bars held

Profit Targets:

- take_profit_r1: ~~23.0R~~ (first partial exit — 25% of position)
- take_profit_r2: ~~45.0R~~ (second partial exit — 25% of position)
- take_profit_r3: 8.0R (final exit — remaining 50%; or 6R/8R for finer scale-out)

Important Kalman Parameters

Kalman Filter:

- `process_noise_1`: 0.01 (default)
- `process_noise_2`: 0.01 (default)
- `measurement_noise`: 500.0 (default)
- `kalman_strength_min`: 70 (default)

Entry Conditions:

- `rsi_min`: 40 (default)
- `rsi_max`: 65 (default) — More conservative than Momentum
- `pullback_percent`: 0.5 (0.5%, default) — Pullback depth
- `stop_loss_pct`: 0.015 (1.5%, default) — Wider stop
- `trailing_stop_pct`: 0.010 (1.0%, default)

Risk Management:

- `risk_per_trade`: 0.015 (1.5%, default)
 - `max_position_pct`: 0.25 (25%, default)
 - `atr_multiplier`: 2.0 (default)
 - `cooldownBars`: 10 (default) — Rest between trades
-

8. RISK MANAGEMENT

Capital Preservation Principles

The most important rule in trading: **Protect Your Capital**

Position Sizing Formula:

Position Size = (Account Balance × Risk %) / (Entry - Stop Loss)

Example:

Account: \$10,000

Risk %: 1%

Entry Price: \$100

Stop Loss: \$98

Distance: \$2

Position Size = ($\$10,000 \times 0.01$) / \$2 = 50 units

Cost: $50 \times \$100 = \$5,000$ (50% of account)

Max Loss: $50 \times \$2 = \100 (exactly 1% of account)

Risk Per Trade Guidelines:

Experience Level	Risk Per Trade	Max Per Month
Beginner	0.5-1.0%	10-15 trades
Intermediate	1.0-1.5%	15-20 trades
Advanced	1.5-2.0%	20-30 trades
Expert	2.0%+	Calculated

DO NOT exceed 2% per trade until proven profitable!

Stop Loss Implementation

Types of Stop Losses:

- 1. ATR-Based Stop (Recommended)** - Uses volatility to set dynamic stops - Formula: $\text{Entry} - (\text{ATR} \times \text{Multiplier})$ - Adapts to market conditions - Used in all built-in strategies
- 2. Percentage Stop** - Fixed percentage below entry - Formula: $\text{Entry} \times (1 - \text{Risk}\%)$ - Simple, easy to calculate - Less adaptive
- 3. Swing-Level Stop** - Below recent swing low - Highest support level - Professional traders love this - Requires manual analysis

Stop Loss Placement Rules:

- ✓ **ALWAYS use stops** - Never trade without one
- ✓ **Place before entering** - Not after
- ✓ **Respect your stops** - Don't move them against you
- ✓ **Use ATR-based** - Better than percentages
- ✓ **Widen in high volatility** - More room for noise

Profit Taking Strategies

Method 1: All-at-Once (Simple)

- Hold full position until exit signal
- Pros: Simple, can catch big moves
- Cons: Gives back profits if reversal

Method 2: Scaling Out (Recommended)

- Exit 1/3 at 1.5R profit
- Exit 1/3 at 2.5R profit
- Exit final 1/3 at 4.0R profit
- Pros: Locks in profits, lets winners run
- Cons: More execution needed

Method 3: Trailing Stop (Good)

- Let position run
- Place trailing stop at 2-5% below high
- Closes automatically if price retraces
- Pros: Locks in profits automatically
- Cons: Can close too early in pullbacks

Recommended: Scaling out (Method 2)

Account Management

Daily Review Checklist:

✓ How many trades taken today? ✓ What was the total P&L? ✓ Which trades were winners/losers? ✓ Did I respect my stops? ✓ Were there obvious missed opportunities?

Weekly Review:

✓ Total trades for the week ✓ Win rate (%) ✓ Average win vs average loss ✓ Biggest trade (good/bad) ✓ Any pattern deviations?

Monthly Review:

✓ Total return (%) ✓ Win rate (%) ✓ Risk-adjusted return (Sharpe) ✓ Max drawdown experienced ✓ Strategy adjustments needed?

9. TECHNICAL INDICATORS REFERENCE

EMAs (Exponential Moving Averages)

What it is: Average price over N periods, weighted toward recent prices

EMA_Fast (Period 5): - Most responsive to recent price - First to trend reversal - Noisy, can whipsaw - Best for: Entry timing

EMA_Mid (Period 26): - Medium-term trend indicator - Balance of responsiveness & stability - Cross with EMA_Slow signals major moves - Best for: Trend confirmation

EMA_Slow (Period 60): - Long-term trend indicator - Stable, less whipsaw - Hard to cross above/below - Best for: Trend direction

How to Read: - **Bullish:** Price > EMA_Fast > EMA_Mid > EMA_Slow - **Bearish:** Price < EMA_Fast < EMA_Mid < EMA_Slow - **Unclear:** EMAs intertwined or out of order

ADX (Average Directional Index)

What it is: Measures strength of trend (not direction)

Range: 0-100

Value	Interpretation	Recommendation
< 20	Weak trend	Don't trade
20-25	Emerging trend	Caution
25-50	Strong trend	Good for entry
> 50	Very strong	Trade with care

How to Read: - **Rising ADX:** Trend getting stronger
- **Falling ADX:** Trend losing strength - **Below 20:** Market is ranging (no trend)

Indicator in Platform: - `adx_min`: 18 (Momentum requires > 18) - `min_adx`: 25 (Kalman prefers > 25)

RSI (Relative Strength Index)

What it is: Measures momentum, overextension

Range: 0-100

Value	Interpretation
< 30	Oversold (too weak, might bounce)
30-70	Neutral (normal trading range)
> 70	Overbought (too strong, might pullback)

How to Read: - **Rising RSI:** Momentum increasing - **Falling RSI:** Momentum decreasing - **Divergence:** RSI makes lower low, price makes higher low (bullish)

In Platform: - `rsi_entry_min`: 40 (don't buy below, too weak) - `rsi_entry_max`: 70 (don't buy above, overbought) - `rsi_exit_threshold`: 75 (exit if RSI > 75)

ATR (Average True Range)

What it is: Measures volatility of price

How it Works: - High ATR = Volatile market (big candles) - Low ATR = Quiet market (small candles) - Adapts to each symbol's characteristics

Used For: - Position sizing (higher volatility = smaller position) - Stop loss placement ($\text{ATR} \times 2.2$ for Momentum) - Entry spacing (higher ATR = need more space)

In Platform: - Stop Loss: Entry - ($\text{ATR} \times 2.2$) - Trailing Stop: Entry - ($\text{ATR} \times 3.0$)

Bollinger Bands

What it is: Price bands showing overbought/oversold levels

Components: - Middle Band: 20-period SMA -
Upper Band: Middle + $2 \times$ standard deviation -
Lower Band: Middle - $2 \times$ standard deviation

How to Read: - **Price near Upper Band:**
Overbought, potential exit - **Price near Lower**
Band: Oversold, potential entry - **Tight Bands:**
Low volatility (ranging market) - **Wide Bands:**
High volatility (trending market)

Ranging Detection: - **Band Width Z-Score** < -0.5:
Bands compressed (ranging) - Combined with
other indicators in 8-check ranging filter

Kalman Filter

What it is: Advanced algorithm that detects trend strength

Output: - **Kalman_Strength:** 0-100, measures strength - **Kalman_Color:** Green (uptrend), Red (downtrend), Gray (neutral) - **Kalman_Oscillator:** Direction and magnitude

How to Use: - **Green color:** Market in uptrend -
Strength > 70: Very strong trend - **Strength < 30:**
Weak, unclear trend - **Color change Red→Green:**
Potential entry on bounce back

In Kalman Strategy: - Entry requires: Green color
AND Strength > 70 - Exit triggered if: Color turns
Red

Volume Indicators

Volume Ratio: - Formula: Current Volume / 20-period Average - **< 1.0:** Light volume (caution) - **1.0-1.3:** Normal volume - **> 1.3:** Heavy volume (good confirmation)

In Platform: - Entry requires: Volume_Ratio > 1.3 - Confirms institutional interest in move

Choppiness Index

What it is: Measures if market is “chopping” (ranging) or trending

Range: 0-100

Value	Interpretation
< 30	Trending (move it)
30-60	Neutral (could go either way)
> 60	Choppy/Ranging (no clear direction)

In Platform: - Ranging detection requires: CHOP > 60 - Combined with 7 other checks for 8-check filter

CCI (Commodity Channel Index)

What it is: Measures deviation from average price

Range: -200 to +200 (extreme values)

Value	Interpretation
< -50	Pullback (undervalued)
-50 to +50	Normal
> +50	Overbought

Pullback Trading: - Entry when CCI < -50 (pullback detected) - Then buys when CCI bounces back above -50 - Kalman strategy uses this logic

MACD (Moving Average Convergence)

Components: - MACD Line (fast) - Signal Line (slow) - Histogram (difference)

How to Read: - **MACD above Signal:** Bullish momentum - **MACD below Signal:** Bearish momentum - **Zero crossover:** Momentum change - **Histogram growing:** Momentum increasing

In Platform: - Exit if: MACD < Signal (momentum loss) - Combined with other signals

Complete Indicator Settings

All indicators used in strategies are fully customizable:

Momentum Strategy Indicators:

EMA Periods: 5, 26, 60

ADX Period: 14 (~~hard~~ minimum: 25 in v7.5 - ~~tr~~
~~end check: 18~~)

RSI Period: 14 (range: 40-70)
CCI Period: 20
ATR Period: 14 (stop multiplier: 2.2)
MACD: 12/26/9
Bollinger Bands: 20/2.0
Kalman: Custom parameters

Kalman Strategy Indicators:

Kalman Filter: Custom tuning
EMA (MA_Fast): 20 periods
EMA (MA_Slow): 50 periods
RSI: 14 periods (range: 40-65)
ATR: 14 periods
ADX: 25 periods
Bollinger Bands: 20/2.0

10. BACKTESTING SYSTEM

Running a Standard Backtest

Step-by-Step:

1. **Select Backtest Mode**
 - Mode dropdown: Choose **Backtest**
 - Confirm "BACKTEST MODE" display
2. **Configure Data Source**
 - Data: **Fetch API Data** (or Use Generated Data)
 - Cache: **✓ Use CSV** (faster)
 - This caches data locally for speed
3. **Set Date Range**

- Start Date: YYYY-MM-DD format
 - End Date: YYYY-MM-DD format
 - Note: More data = longer backtest
 - 4. **Choose Strategy**
 - Strategy Type: **Momentum, Kalman, or Enhanced**
 - Parameters: Default or Custom
 - 5. **Select Backtest Type**
 - **Standard Backtest:** Single run with your params
 - **Optimization:** Test 100+ param combinations
 - 6. **Optional: Monte Carlo**
 - ☒ Enable Monte Carlo
 - Simulations: 1000 (default)
 - Trades per sim: 100
 - Method: Hybrid (recommended)
 - 7. **Click  Run Backtest**
 - Watch progress in log
 - Results appear when complete
-

Understanding Backtest Results

Key Metrics Explained:

Final Equity: Total account value at end - Example: Started with \$50,000, ended with \$75,432 - Profit: \$25,432 (50.8% return)

Total Trades: Number of closed trades - Only counts COMPLETED trades (entries + exits) - Pending positions don't count

Win Rate (%): Percentage of winning trades - Formula: $(\text{Winning Trades} / \text{Total Trades}) \times 100$ - Example: 25 wins out of 45 trades = 55.6% - Target: > 50% is good, > 55% is excellent

Return (%): Total percentage gain - Formula: $((\text{Final Equity} - \text{Initial}) / \text{Initial}) \times 100$ - Example: \$75,432 / \$50,000 = 150% growth

Sharpe Ratio: Risk-adjusted returns - Measures return per unit of risk taken - Formula: Average Return / Standard Deviation - > 1.0 is good, > 1.5 is excellent - Ignores luck, shows true consistency

Sortino Ratio: Like Sharpe but only counts downside risk - Better metric than Sharpe for trading - Ignores upside volatility (good volatility) - > 1.5 is good, > 2.0 is excellent

Max Drawdown (%): Biggest loss from peak to trough - Example: Peak \$75k → Trough \$65k = 13% drawdown - Measures pain endured - Target: < 20%, better < 15%

Best Trade (%): Single largest winning trade - Example: +4.5% gain on one trade - Shows upside potential

Worst Trade (%): Single largest losing trade - Example: -2.1% loss on one trade - Shows downside protection

Avg Trade (%): Average gain/loss per trade -
Formula: Total P&L / Number of Trades - Positive = profitable, negative = losing

Profit Factor: Total wins / Total losses - Formula:
Sum of Winning Trades / Sum of Losing Trades - > 2.0 is good (2:1 win-to-loss ratio) - > 3.0 is excellent (3:1 ratio)

Evaluating Backtest Quality

GOOD Backtest Results:

✓ Win Rate 50-60% ✓ Sharpe 1.0-2.0 ✓ Max Drawdown < 20% ✓ Profit Factor > 2.0 ✓ 30-100 trades minimum ✓ Consistent monthly returns ✓ Avg Trade > 0.5%

RED FLAGS (Don't Trade This):

✗ Win Rate < 45% ✗ Sharpe < 0.8 ✗ Max Drawdown > 30% ✗ Profit Factor < 1.5 ✗ Only 5-10 trades (too few) ✗ All profits from 1-2 trades ✗ Negative average trade

Questions to Ask:

Is this real? - Realistic commission (0.1% per trade)? - Realistic slippage (0.1-0.2%)? - Not using lookahead bias?

Will it work live? - Good historical performance? -
Works in multiple market periods? - Consistent
month-to-month?

Can I trade it? - Enough trades per month (8-15)? -
Small enough position sizes? - Stop losses realistic?
- Can I follow the signals?

Parameter Optimization

What is Optimization?

Instead of manually testing each parameter combination, the system automatically tests 100+ combinations and finds the best:

Testing EMA Periods:

- 5/26/50 (slow)
- 5/26/60 (default)
- 5/26/70 (faster)
- ... 100+ more combinations

Scoring Each:

- Sharpe ratio
- Return %
- Win rate
- Risk-adjusted metrics

Finding Best:

Winner: EMA 5/26/75 with Sharpe 1.8

Running Optimization:

1. **Set up backtest** (as above)
2. **Select Backtest Type: Optimization**

3. Click **Run Backtest**

- Takes 2-5 minutes for 100 combinations
- Watch log for progress

4. **Review Results:**

- Best parameters shown
- Before/after improvement shown
- Optimization impact metrics

Important Notes:

 **Curve Fitting Risk:** - Optimization finds parameters that worked GREAT in the past - Might FAIL in the future (overfitting) - Solution: Test on out-of-sample data - Solution: Use common-sense constraints

 **Best Practices:** - Optimize on 1-2 years minimum - Test results on recent data (last 3 months) - Constrain parameters (don't let them go wild) - Look for ROBUST optimization (small changes = small impact)

Excel Report Analysis

Sheet 1: Market Data

Contains all OHLCV + indicators: - Open, High, Low, Close, Volume - EMA_Fast, EMA_Mid, EMA_Slow - ADX, RSI, ATR - Bollinger Bands, MACD - 20+ more indicators

Use For: - Visual analysis of price moves - Indicator confirmation - Finding best times to trade

Sheet 2: Trades

Details every trade: - Entry Time, Entry Price - Exit Time, Exit Price - Quantity, P&L, Return % - Duration, Exit Reason - Confluence Score (0-7)

Column Breakdown:

Trade #: Sequential trade number (1, 2, 3...)

Entry Time: Exact date/time of buy signal

Entry Price: Price you bought at

Exit Time: Exact date/time of exit signal

Exit Price: Price you sold at

Exit Reason: Why the trade closed: - stop_loss: Hit stop loss - profit_target_x: Reached target - trend_reversal: Trend changed - time_exit: Held too long - momentum_loss: MACD turned negative - ema_cross: EMAs crossed down

P&L (\$): Profit or loss in dollars - Positive = winner - Negative = loser

Return (%): Percentage return - Formula: $(\text{Exit} - \text{Entry}) / \text{Entry} \times 100$ - Example: $(101 - 100) / 100 \times 100 = 1\%$

Duration: How long you held it - In bars (1h interval = 1 bar per hour) - Or in hours/days

Win/Loss: Y/N indicator

Confluence: Quality score 0-7 - 5: Minimum filters met - 6: +1 additional confirmation - 7: +2 additional confirmations

Use For: - Identify best/worst trades - Find exit reason patterns - Learn from mistakes - Track trade quality over time

Sheet 3: Summary

High-level statistics: - Total Return % - Sharpe Ratio - Win Rate % - Profit Factor - Best/Worst Trades - Monthly Breakdown - Risk Metrics

Use For: - Quick performance overview - Comparing strategies - Sharing with others

11. MONTE CARLO SIMULATION

What is Monte Carlo?

Monte Carlo simulation uses probability to estimate what could happen in the future based on past performance.

Example: - Your backtest had 50 trades, won 55% - Monte Carlo runs 1000 simulations of “50 random trades from your history” - Shows the range of

possible outcomes - Gives confidence intervals
(worst/best case scenarios)

Running Monte Carlo

Prerequisites:

- Complete a standard backtest first
- ☒ Enable Monte Carlo in backtest settings

Configuration:

Simulations: 100-5000 - 100-500: Quick estimate -
1000-2000: Balanced (recommended) - 5000:
Maximum accuracy (slow)

Trades per Simulation: 50-500 - Should match
your typical monthly trades - 100 is good default

Method: Parametric, Bootstrap, or Hybrid -

Parametric: Assumes normal distribution (fastest)

- **Bootstrap:** Resamples actual trades (most

realistic) - **Hybrid:** Combines both (recommended)

Running:

1. **Configure backtest** (as above)
 2. ☒ **Enable Monte Carlo**
 3. **Set parameters** (simulations, trades,
method)
 4. **Click ☒ Run Backtest**
 5. **Wait for completion** (takes 30 seconds - 5
minutes)
-

Understanding Monte Carlo Results

Output Includes:

Confidence Intervals:

Median Outcome:	\$67,500 (50% chance)
Best Case (95%):	\$92,000 (best 5%)
Good Case (75%):	\$85,000 (best 25%)
Poor Case (25%):	\$48,000 (worst 25%)
Worst Case (5%):	\$32,000 (worst 5%)

Interpretation: - If you run the strategy for 100 more trades - 95% confidence your account will be between \$32k-\$92k - Most likely outcome around \$67,500

Probability Metrics:

Profit Probability: 87% - 87% chance you'll make money - 13% chance you'll lose money - Good strategy if > 60%

Maximum Drawdown Scenarios: - 5% chance of losing 25% - 25% chance of losing 15% - 50% chance of losing < 10%

Using Monte Carlo Results

Decision Framework:

If Profit Probability > 80%: ☒ High confidence strategy ☒ Proceed to live trading ☒ Use normal position sizing

If Profit Probability 60-80%: ⚠ Moderate confidence ⚠ Maybe reduce position size 30-50%
⚠ Trade smaller amounts first

If Profit Probability < 60%: ✗ Low confidence
✗ Don't trade this live ✗ Optimize parameters more

Risk Considerations:

Drawdown Tolerance: - Can you lose 25% (worst case)? - If not: Reduce position sizes - If yes: You can handle normal strategy

Capital Requirements: - Worst case shows minimum needed - Always keep 2-3x this buffer - Example: Worst case \$32k → Keep \$90k+ account

12. MACHINE LEARNING INTEGRATION

Introduction to ML

Machine learning teaches computers to recognize patterns in data without explicit programming.

For Trading: - ML learns patterns from past price data - Predicts future price direction - Assigns confidence to predictions - Can supplement traditional analysis

Three ML Models Available

1. Random Forest Model

How it works: - Creates many decision trees - Each tree votes on prediction - Final prediction = majority vote - Fast and robust

Speed: ⚡ ⚡ ⚡ Fastest **Accuracy:** ★ ★ ★ 75-80% **Training Time:** 30 seconds **Best For:** Real-time trading, quick decisions

Pros: ✅ Very fast predictions ✅ Handles nonlinear relationships ✅ Robust to outliers ✅ Can run on slower computers

Cons: ❌ Less accurate than XGBoost ❌ Doesn't learn complex patterns as well

2. XGBoost Model

How it works: - Advanced gradient boosting algorithm - Trees sequentially improve predictions - Each tree corrects previous errors - Powerful and accurate

Speed: ⚡ ⚡ Medium **Accuracy:** ★ ★ ★ ★ 80-90% **Training Time:** 1-2 minutes **Best For:** Accurate predictions, research

Pros: ✅ Highest accuracy ✅ Handles complex patterns ✅ Better than Random Forest ✅ Industry standard

Cons: ❌ Slower predictions ❌ Needs more compute ❌ Slightly longer training

3. LSTM Neural Network

How it works: - Deep learning (artificial neural network) - Remembers long-term patterns - Learns temporal dependencies - Can forecast multiple candles

Speed: ⚡ Slowest **Accuracy:** ★★★★★
85%+ **Training Time:** 2-5 minutes **Best For:**
Pattern recognition, forecasting

Pros: ✅ Highest accuracy potential ✅ Can forecast multiple bars ahead ✅ Learns temporal patterns ✅ Most sophisticated

Cons: ❌ Slowest predictions ❌ Requires significant compute ❌ Longest training time ❌ Less interpretable ("black box")

Using ML in Trading

Enabling ML:

1. **Demo/Live/Backtest Mode:** Any mode works
2. **Click ☒ Enable ML**
3. **Select Model:**
 - Random Forest (start here)
 - XGBoost (recommended)
 - LSTM (advanced)

4. **Model trains automatically** (30 seconds - 5 minutes)
5. **Watch log:** "Model trained successfully!"

Confidence Threshold:

The platform shows ML confidence 0-100%:

ML Prediction: BULLISH (Confidence: 78%)

Threshold Setting: - Default: 75% (only trade if confidence $\geq$ 75%) - Range: 55-95% - Higher = fewer, higher-confidence signals - Lower = more signals, lower quality

Recommendation: - Start with 75% - Adjust based on win rate

Auto-Execute Option:

 **Enable Auto-Execute ML:** Trade automatically on signals

 **CAUTION:** - Only enable if you trust the model - Start with small position sizes - Monitor first 10-20 trades manually - Turn off if performance degrades

ML Prediction Confidence

How to Interpret:

90-100%: Very high confidence - Strongest possible signal - Trade full size if comfortable - Expect good hit rate

75-90%: High confidence - Good signal, trade normally - This is the default threshold - Good entry point

60-75%: Medium confidence - Decent signal, maybe reduce size - Below default threshold - Consider other confirmation

40-60%: Low confidence - Model uncertain - Usually filtered out - Skip this trade

< 40%: Model confused - No clear pattern - Market too uncertain - Wait for better setup

Multi-Candle Forecasting

The platform can forecast 1-20 candles ahead:

Forecast Candles: 5

ML Prediction for next 5 candles:

Candle 1: BULLISH (up)

Candle 2: BULLISH (up)

Candle 3: NEUTRAL (uncertain)

Candle 4: BEARISH (down)

Candle 5: BEARISH (down)

Use For: - See predicted trend direction - Plan exit points ahead of time - Identify reversal patterns - Set profit targets intelligently

Note: Accuracy decreases further out - Candle 1: Most accurate - Candle 5: Less accurate - Use as guide, not gospel

Combining ML with Technical Analysis

Strategy:

1. **Technical Analysis:** Does EMA alignment look good?
2. **ML Confirmation:** Does model agree (confidence > 75%)?
3. **Both Agree?** → Very strong signal
4. **Disagree?** → Be cautious, smaller position

Example:

Technical Signal:

- EMA alignment: YES (bullish)
- ADX: 28 (strong)
- RSI: 55 (neutral)
- Volume: 1.4x average (strong)
- Signal Quality: Good

ML Signal:

- Model: BULLISH
- Confidence: 82%
- Forecast: UP,UP,NEUTRAL,DOWN
- Signal Quality: Good

Result: STRONG AGREEMENT → Full-size trade

Example 2:

Technical Signal:

- EMA alignment: YES (bullish)
- ADX: 28 (strong)

- Signal Quality: Good

ML Signal:

- Model: BEARISH
- Confidence: 45%
- Signal Quality: Weak (below threshold)

Result: DISAGREEMENT → Skip this trade (or reduce size)

ML Model Performance

Expected Accuracy:

Random Forest: 75-80% - Prediction: BULLISH - Actual: BULLISH ✓ - Actual: BEARISH ✗ - ~75-80 correct out of 100

XGBoost: 80-90% - More accurate than Random Forest - ~80-90 correct out of 100

LSTM: 85%+ - Highest accuracy - ~85+ correct out of 100

In Trading Context:

Win Rate < Model Accuracy - ML accuracy 85% ≠ Trading win rate 85% - Trading also depends on: - Position sizing - Stop losses (can hit false signals) - Market conditions - Risk management - Entry timing

Expected Trading Win Rate: - ML accuracy 85% -
With good risk mgmt: 50-60% actual win rate -
With excellent risk mgmt: 60-70%

13. PARAMETER MANAGEMENT

Default vs Custom Parameters

DEFAULT PARAMETERS:

✅ Pre-optimized by developers ✅ Proven to work across many markets ✅ Conservative approach ✅ Recommended for most traders ✅ Can't be modified during trading

CUSTOM PARAMETERS:

⚠️ User-defined settings ⚠️ Allows fine-tuning ⚠️ Can optimize for specific markets ⚠️ Risk of curve-fitting ⚠️ Requires good understanding

Switching Modes

To Use Default:

1. **Open Settings** (⚙ button)
2. **Go to Parameter Tab**
3. **Select:** "Default Parameters"
4. **Close Settings**
5. Default values now loaded

To Use Custom:

1. **Open Settings** (⚙ button)
 2. **Go to Parameter Tab**
 3. **Select:** "Custom Parameters"
 4. **Modify values** as desired
 5. Click  **Save Custom Parameters**
 6. Custom values now loaded
-

Momentum Parameter Guide

EMA Configuration

ema_fast_period: Default 5

How it affects: - Smaller value (~~3-5-8~~): More responsive, more false signals - Larger value (~~7-10-15~~): Less responsive, fewer signals - v7.5 default: 9 (optimised for 15m crypto)

Adjustment Guide: - More entries wanted? Lower to ~~7-8-4~~ - Too many false signals? Raise to ~~12-15-10~~ - Default 9 (v7.5) is balanced

ema_mid_period: Default 21 (v7.5: changed from 26)

- Smaller (15-20): Faster trend detection
- Larger (30-40): Slower, more stable

ema_slow_period: Default 50 (v7.5: changed from 60)

- Smaller (45-50): Responds faster
- Larger (75-100): More stable

Important: Always keep EMA_Fast < EMA_Mid < EMA_Slow!

ADX Filter

adx_min: Default 25 (v7.5 hard min)~~18~~

Value	Effect
10	Too loose, trades in weak trends (<u>not recommended</u>)
18	<u>Below v7.5 hard minimum (avoid)</u> Default, good filter
25	<u>v7.5 hard minimum — Default</u> Tight, only strong trends
30+	Very tight, few trades

Adjustment Guide: - Want more trades? Lower to 22 (do NOT go below 20 — choppy stops)~~15~~ - Too many losing trades? Raise to 28-30~~22-25~~ - Default 25 (v7.5 hard minimum)~~18 is balanced~~

RSI Filter

rsi_entry_min: Default 40 **rsi_entry_max:** Default 70

Min	Max	Effect
30	75	Loose, more trades
40	70	Default, balanced
50	65	Tight, fewer trades

Adjustment Guide: - Too few entries? Widen to 40-65~~range (30-75)~~ - Too many entries? Tighten to 48-56~~range (45-65)~~ - Default (45-58~~40-70~~) is the v7.5 pullback zone~~balanced~~

Risk Management

risk_full_position: Default 0.015 (1.5% per trade)

Value	Effect
0.5%	Very conservative
1%	Conservative
1.5%	Default (balanced)
2%	Aggressive
2.5%+	Very aggressive

Adjustment Guide: - Tier 1 (score 75-87): use 1.2% - Tier 2 (score ≥88): use 1.5%~~Want slower growth? Use 0.5-1% - Need faster growth? Use 1.5-2%~~ - **DON'T exceed 2%** until proven profitable

stop_loss_atr_mult: Default 2.2

- Lower (1.8-2.0): Tighter stops, more stopped out
- Default (2.2): Balanced
- Higher (2.5-3.0): Wider stops, bigger losses

Adjustment Guide: - Getting stopped out too much? Raise to 3.5x~~2.5~~ - Losses too big? Lower to 2.5x - Default 3.0x (v7.5: widened from 2.2x)~~0~~

Kalman Parameter Guide

Kalman Filter Settings

kalman_strength_min: Default 70

- Lower (50-60): More entries
- Default (70): Balanced
- Higher (80-90): Fewer entries

Adjustment: Want more trades? Lower to 60

Entry Risk

stop_loss_pct: Default 1.5%

Kalman uses FIXED percentage stops (not ATR): -
1.0%: Tight stop, higher frequency - 1.5%: Default, balanced - 2.0%: Wide stop, fewer stops

Adjustment: Too many stops? Raise to 2.0%

trailing_stop_pct: Default 1.0%

- How much price can pull back before closing
- Lower (0.5-1.0%): Tight, exits early
- Higher (1.5-2.0%): Loose, lets winners run

Adjustment: Giving back profits? Lower to 0.5-0.75%

Saving Your Configurations

Automatic Saving:

When you click “ Save Custom Parameters”: -
Settings saved to strategy_settings.json -
Persists between sessions - Automatically loaded
on startup

Reset to Defaults:

Button “ Reset to Defaults”: - Restores original
values - Removes custom parameters - Can't undo!

14. ADVANCED FEATURES

Quality Score System~~Confluence Scoring~~ (0- 100~~7~~ Scale)

Measures entry quality:

Score $\geq$ 88/100 (Tier 2 – Exceptional): - ADX 30-34 (Strong trend, 90% weight) - RSI 45-58 (Pullback zone, full points) - MACD line > signal + histogram rising - Volume $\geq$ 2.0x average - EMA: Close > EMA9 > EMA21 > EMA50 (Perfect stack) - Risk: 1.5% per trade | Threshold raised 85→88 in v7.5
Score 75-87/100 (Tier 1 – Good): - Mandatory filters pass (ADX $\geq$ 25, RSI 45-58, Vol $\geq$ 1.2x) - MACD above signal, daily trend up - Risk: 1.2% per trade
Score < 75/100 (Rejected): - Quali

ty too low – no trade entered - Scores 60-74 logged as near-misses for fuzzy learning

~~Score 5:~~

- ~~– Only essential filters met~~
- ~~– Minimum acceptable signal~~
- ~~– Trade with smaller size~~

~~Score 6:~~

- ~~– Essential + 1 added confirmation~~
- ~~– Good signal~~
- ~~– Trade normal size~~

~~Score 7:~~

- ~~– Essential + 2 added confirmations~~
- ~~– Best possible signal~~
- ~~– Can trade larger size~~

Excel Export: QualityConfluence score (0-100) and tier in trades sheet

Use For: - Identify best/worst quality entries -
Learn from mistakes - Size positions by quality

Exit Reason Tracking (20+ Categories)

Platform tracks WHY each trade closed:

stop_loss: Hit stop loss (forced out)
trailing_stop: Trailing stop triggered
profit_target_1r: First profit level
profit_target_2r: Second profit level
profit_target_3r: Final profit level
trend_reversal: EMA crossover detected
momentum_loss: MACD turned negative

overbought_rsi: RSI > 75
ema_cross: EMAs crossed down
time_exit: Max hold time reached
breakeven: Closed at entry price
ml_exit_confirmed: ML signal + technical confirm

Analyze Your Exits: - Which reasons are most profitable? - Which reasons cause losses? - Are you exiting too early? - Are your stops working?

In Excel: “Exit_Reason” column in Trades sheet

Equity Curve Analysis

What it shows: How your account balance changes trade-by-trade

Example:

Start: \$50,000
Trade 1: +\$500 → \$50,500 (Win)
Trade 2: -\$400 → \$50,100 (Loss)
Trade 3: +\$800 → \$50,900 (Win)
...
End: \$75,432

Equity Curve = [50000, 50500, 50100, 50900
, ... 75432]

Good Equity Curve:  Generally trending up 
Smooth, steady growth  Manageable drawdowns
 Recovers quickly from losses

Bad Equity Curve: ✗ Flat or declining ✗ Huge swings ✗ Slow recovery from losses ✗
“Sawtooth” pattern (up/down/up/down)

Visual: Charts show equity curve automatically

Volume Strength Gauge

Real-time visual indicator showing market sentiment:

RED	←	NEUTRAL	→	GREEN
Selling		BALANCED		Buying Pressure
<-50		0-0		+50>

Red (Negative): More selling than buying - Caution on long entries - Good for short entries

Green (Positive): More buying than selling - Good for long entries - Caution on short entries

Use For: - Quick market sentiment check -
Confluence with technical signals - Volume confirmation

Parameter Constraints

The platform enforces constraints to prevent invalid configurations:

EMA Order: - EMA_Fast MUST be < EMA_Mid -
EMA_Mid MUST be < EMA_Slow - If violated:
Parameter rejected

ADX Minimum: - Can't set below 20 (causes choppy ATR-wide stops in v7.5) - Hard minimum 2510 (too loose) - Can't set above 50 (too tight)

RSI Range: - Min < Max always - Min ≥ 0 , Max ≤ 100

Risk Per Trade: - Minimum 0.1% - Maximum 5% - Recommended: 0.5-2%

Position Tracking

Platform tracks open positions in real-time:

Displays: - Entry price - Current price - Unrealized P&L - Bars held - Current status

Updates: - Every candle (your timeframe) - Real-time (for live trading) - Historical (for backtests)

15. TROUBLESHOOTING

Common Issues & Solutions

“Connection Failed” Error

Problem: API connection not working

Solutions: 1. Check config.json exists with correct credentials 2. Verify API key is correct (copy-paste from OKX) 3. Check internet connection 4. Try “Check Connection” again 5.

Verify API permissions on OKX 6. Try different API key

If Still Failing: - Use Demo mode while troubleshooting - Contact support with error message

“Insufficient Balance” Error

Problem: Not enough money for trade

For Demo Mode: - Doesn't happen, has unlimited virtual balance - Check order size % is reasonable

For Live Mode: - Account balance too low for order size - Solution 1: Add more capital to account - Solution 2: Lower order size % - Solution 3: Trade smaller symbol

Example: - Account: \$500 - Order Size: 30% - Attempting to trade: \$150 - At \$100 entry = 1.5 units - If ATR = \$5 and stop 2.2x = \$11 stop - Then $\$150 / \$11 = 13.6$ units - Works fine!

“No Data Available” Error

Problem: Backtest can't find historical data

Causes: 1. Invalid date range 2. Too short timeframe 3. Symbol spelling wrong 4. API connection issue

Solutions: 1. Check date format: YYYY-MM-DD 2. Ensure at least 100 days of data 3. Verify symbol: SOL-USDT (with dash) 4. Try “Check Connection” first 5. Try with “Use Generated Data” to test

“Stop Loss Not Triggering” Error

Problem: Stop loss level not being used

Causes: 1. Stop loss level too wide 2. Stop loss level above entry price 3. Position size calculation error

Solutions: 1. Check ATR is calculated correctly 2. Verify: Entry > Stop Loss (for long) 3. Check position size is positive 4. Review logs for stop loss price shown

Backtest “No Trades” Result

Problem: Strategy never enters a trade

Causes: 1. Filters too tight (ADX, RSI, etc.) 2. Date range has no qualifying signals 3. Parameters don’t match data 4. Ranging detection too sensitive

Solutions: 1. LowerLoosen ADX from 25 to 22 (do NOT go below 20 in v7.5(try 15 instead of 18)) 2. Widen RSI range (try 40-65 instead of 45-5830-75) 3. Disable ranging detection 4. Backtest different date range 5. Check data loaded correctly

Poor Backtest Results

Problem: Strategy underperforms expectations

Causes: 1. Bad parameters 2. Bad market condition for strategy 3. Symbol doesn't fit strategy 4. Filters too tight 5. Time period unlucky

Solutions: 1. Run Optimization to find better params 2. Test on different time period 3. Adjust entry filters looser 4. Try different symbol (different characteristics) 5. Increase sample size (more trades)

ML Model Won't Train

Problem: "ML model training failed"

Causes: 1. Not enough data (< 200 candles) 2. Data corrupted 3. Model parameters invalid 4. Insufficient memory

Solutions: 1. Run backtest first to load data 2. Get at least 300 candles of data 3. Restart the application 4. Try Random Forest instead (lighter) 5. Disable other memory-intensive apps

Backtest Takes Too Long

Problem: Backtest running for 30+ minutes

Causes: 1. Too large date range 2. Optimization running (many combinations) 3. Monte Carlo with high simulations 4. Slow computer

Solutions: 1. Reduce date range (1 year instead of 5) 2. Standard backtest instead of optimization 3. Reduce Monte Carlo to 100 sims 4. Close other applications 5. Try overnight if running anyway

Excel Export Not Working

Problem: Can't open Excel file or file corrupted

Causes: 1. File still open from previous run 2. Excel not installed 3. Insufficient disk space 4. File permissions issue

Solutions: 1. Close all Excel windows 2. Reinstall openpyxl: `pip install --upgrade openpyxl` 3. Check disk space (need 100MB free) 4. Run as administrator 5. Try saving to different location

General Troubleshooting Steps

1. Restart Application

- Close completely
- Open again
- Fixes most issues

2. Verify Connection

- Click "Check Connection"
- Should show "Connection successful!"

- If not, check config.json
 - 3. **Check Logs**
 - Read the Trading Log (bottom section)
 - Look for error messages
 - Note exact wording
 - 4. **Test with Demo**
 - Switch to Demo mode
 - See if issue reproduces
 - Narrows down problem
 - 5. **Check Data**
 - Backtest with same settings
 - If backtest works, issue is live mode
 - If backtest fails, issue is data/parameters
 - 6. **Simplify Setup**
 - Disable ML
 - Disable Monte Carlo
 - Use default parameters
 - See if issue persists
 - 7. **Get Help**
 - Note exact error message
 - Describe what you were doing
 - Share relevant logs
 - Contact support
-

16. BEST PRACTICES

Daily Trading Routine

Before Market Opens:

1. **Check Technical Setup**
 - Any major overnight news?
 - Are daily EMAs still aligned?
 - What's the trend direction?
2. **Review Settings**
 - Correct symbol selected?
 - Correct timeframe?
 - Strategy loaded?
 - ML enabled? (if desired)
3. **Test Connection**
 - Click "Check Connection"
 - Should be green ✓
4. **Start Platform**
 - Click "Start Trading"
 - Watch first 30 min closely

During Trading:

1. **Monitor Signals**
 - Watch for entry signals
 - Don't force trades
 - Let setup come to you
2. **Manage Positions**
 - Track P&L
 - Don't panic if small loss
 - Respect your stops

3. **Note Observations**

- What's working?
- What's not?
- Any patterns?

End of Day:

1. **Review Trades**

- How many trades?
- What was the P&L?
- Any obvious mistakes?

2. **Check Open Positions**

- Any positions still open?
- Is stop loss reasonable?
- Review holding time

3. **Plan Tomorrow**

- Any parameter adjustments needed?
 - Market conditions changing?
 - Strategy still fitting?
-

Weekly Review

Every Friday:

1. **Review Statistics**

- Total trades: 8-15 is typical
- Win rate: 50-60% target
- Average P&L: Should be positive

2. **Analyze Trades**

- Best trade: Why was it so good?
- Worst trade: What went wrong?
- Pattern: Any obvious problems?

3. **Strategy Assessment**
 - Is Momentum still working?
 - Should I switch to Kalman?
 - Need to optimize parameters?
 4. **Risk Check**
 - Largest loss: Too big?
 - Largest win: Did I size correctly?
 - Account: Stable growth?
 5. **Plan Next Week**
 - Any adjustments?
 - Try different symbol?
 - Tighten or loosen filters?
-

Monthly Review

Last Day of Month:

1. **Full Performance Review**
 - Total return: Should be positive
 - Win rate: Track vs. target
 - Sharpe ratio: Consistency check
 - Max drawdown: Risk assessment
2. **Strategy Effectiveness**
 - Which strategy performed best?
 - Momentum or Kalman better?
 - Market conditions favorable?
3. **Learning Moments**
 - Big wins: What did right?
 - Big losses: What went wrong?
 - Key insights: Lessons learned?

4. **Optimization Decision**

- Should I backtest for optimization?
- Are parameters still valid?
- Market changed significantly?

5. **Next Month Planning**

- Same strategy or change?
 - Adjust parameters?
 - More aggressive or conservative?
 - Increase position size or reduce?
-

Risk Management Best Practices

GOLDEN RULES:

✅ **Rule 1: Never Risk > 2% Per Trade** - Hard limit - Better: 0.5-1.5% - Exception: Expert only

✅ **Rule 2: Always Use Stop Losses** - Never trade naked - Mental stops don't count - Hard stops only

✅ **Rule 3: Let Winners Run** - Don't exit too early - Use trailing stops - Scale out, don't exit all

✅ **Rule 4: Cut Losses Quickly** - If setup breaks, exit immediately - Don't hope for recovery - Accept small loss, protect capital

✅ **Rule 5: Journal Your Trades** - Track every entry/exit - Note what worked - Learn from mistakes

✅ **Rule 6: Review Regularly** - Daily: Quick check
- Weekly: Detailed review - Monthly: Strategy
assessment

Parameter Optimization Best Practices

DO:

✅ Optimize on 1-2 years minimum ✅ Use
realistic constraints ✅ Test on recent data (3-6
months) ✅ Look for ROBUST improvements ✅
Trade slightly MORE conservative than backtest ✅
Re-optimize quarterly

DON'T:

❌ Optimize on just 3 months ❌ Remove all
constraints ❌ Change only one parameter at a
time ❌ Use optimization results without
validation ❌ Trade immediately after
optimization ❌ Assume optimization will work
forever

Position Sizing Framework

Conservative (Recommended Start):

- Risk: 0.5% per trade
- Position size: Small
- Growth: Slow (20-40% annual)
- Stress: Low

- Beginner:  YES

Moderate (After Proven):

- Risk: 1-1.5% per trade
- Position size: Medium
- Growth: Good (60-100% annual)
- Stress: Moderate
- Intermediate:  YES

Aggressive (Expert Only):

- Risk: 1.5-2% per trade
- Position size: Large
- Growth: Fast (100%+ annual)
- Stress: High
- Advanced:  YES

Overly Aggressive (NOT RECOMMENDED):

- Risk: > 2% per trade
- Position size: Very large
- Growth: Very fast
- Stress: Extreme
- Blowup: Highly likely

Recommendation: Start Conservative, move up as proven

17. FAQ - FREQUENTLY ASKED QUESTIONS

General Questions

Q: How much money do I need to start? A: - Demo: \$0 (virtual \$1,000) - Backtest: \$0 - Live: Minimum \$500-\$1,000 recommended - Starting smaller is actually better (less risk)

Q: Is past performance a guarantee? A: No. All disclaimers apply. - Backtest results don't guarantee future performance - Market conditions change - Strategy might underperform - Always assume risk of loss

Q: How long before I'm profitable? A: Highly variable: - Demo: Immediately if you learn well - Live: Could take months to find edge - Typical: 3-6 months of learning - Many never become profitable (be honest)

Q: Can I trade with \$100? A: Technically yes, but: - Very small positions only - High % risk per trade - Hard to manage properly - Recommend \$500+ minimum

Strategy Questions

Q: Which strategy should I use? A: Depends on your situation: - **Momentum:** Trending markets, active trader - **Kalman:** Mean-reversion, less

monitoring - **Enhanced:** Custom needs, advanced trader - Start with Momentum if unsure

Q: Why doesn't Momentum work in ranging markets? A: Because: - Momentum chases price up/down - Ranges mean price bounces back - You chase rallies that reverse - Solution: 8-check ranging filter disables trades

Q: Can I use both strategies at once? A: Not directly, but: - Run two separate instances - Different timeframes per strategy - Backtest separately - Compare results

Q: How often should I optimize parameters? A: Typically: - Initial optimization: Before you start - Re-optimize: Every 3 months - After major market change - After significant drawdown

Risk Management Questions

Q: Should I risk 2% or 1% per trade? A: - Conservative traders: 0.5-1% - Moderate traders: 1-1.5% - Aggressive traders: 1.5-2% - Never above 2% - Start with 0.5-1%

Q: What if I hit my max daily loss? A: Take the day off: - Stop trading for the day - Don't revenge trade - Review what went wrong - Take emotional break - Trade again tomorrow

Q: How do I size positions correctly? A: Use the formula:

Position Size = (Account × Risk %) / (Entry - Stop Loss)

Example:

Account = \$5,000

Risk = 1%

Entry = \$100

Stop = \$98

Position = (5000 × 0.01) / 2 = 25 units

ML Questions

Q: Should I use ML? A: It depends: -  Use if:

Want higher accuracy, comfortable with tech - 

Skip if: Prefer simplicity, already profitable -

Recommended: Turn on after 100+ manual trades -

Not required for profitability

Q: Which ML model is best? A: - Random Forest:

Fast, good starting point - XGBoost: Best accuracy,

recommended - LSTM: Highest potential, but

slower - Try XGBoost first

Q: Why does ML give confident wrong

predictions? A: Because: - Markets have

randomness - No model is 100% accurate - Past

patterns don't guarantee future - Always combine

with technical analysis

Backtest Questions

Q: Why is backtest profitable but I'm losing live?

A: Common reasons: 1. Slippage: Backtest underestimates spread 2. Execution: Can't get exact backtest prices 3. Emotions: Living trading psychology different 4. Curve-fitting: Parameters overfit past data 5. Time: Live trading still statistically insignificant

Q: How long should I backtest? A: - Minimum: 1 year - Recommended: 2-3 years - Better: 5+ years - More data = more reliable

Q: Should I optimize? A: Yes, but carefully: - Optimize on 2+ years - Test on separate recent 6 months - Look for ROBUST improvements - Don't curve-fit

Q: My backtest has 1000 trades, is that enough?

A: That's actually suspicious: - Probably optimized too much - Might be curve-fitting - Test on completely new data - Fewer, genuine trades better

Trading Questions

Q: When is the best time to trade? A: Generally: - During active market hours - Not during major news - Not overnight gaps - Crypto: 24/5 but some better times - Test with backtesting

Q: Should I ever override the signal? A: Generally no: - Trust your system (you proved it works) -

Emotions override logic - Exception: Major news event - Let system handle normal trading

Q: What if I miss a signal? A: Don't chase: - Move on to next signal - Don't FOMO entry - Worst thing you can do - Be patient, another signal coming

Q: Should I add to winning trades? A: Carefully: - NOT recommended for beginners - Only if strong second signal - Use smaller size than original - Don't increase risk per trade

Troubleshooting Questions

Q: Why isn't my strategy trading? A: Most common reasons: 1. Ranging market (8-check filter blocks it) 2. ADX too low (no trend) 3. Wrong market conditions 4. Filters too tight 5. Market structure changed

Q: Why is my win rate lower than backtest? A: - Slippage (prices worse than backtest) - Gaps (market gap against you) - Execution timing - Still early (need more statistical sample) - Market has changed

Q: Should I change my stops mid-trade? A: NO: - Hard rule: Never move stops against you - Moving stop losses widens = bigger losses - Violates risk management - Discipline is key

Advanced Questions

Q: Can I run multiple timeframes

simultaneously? A: Yes: - Run separate instances - Different symbol + timeframe each - Or: Same symbol, different timeframe - Monitor carefully to avoid over-leverage

Q: How do I handle black swan events?

A: You can't, but: - Good risk management helps - Emergency stop button ready - Accept large moves happen - Wider stops help - Position sizing is key

Q: What's the maximum annual return possible?

A: Theoretically unlimited, but realistically: - Conservative: 30-50% annual - Moderate: 60-100% annual - Aggressive: 100-200%+ annual - Beyond: Increasingly unrealistic - Most traders: 20-40% annual

Closing Questions

Q: How do I know if I'm ready to trade live?

A: You should have: -  200+ demo trades - 
Consistent win rate > 50% -  Positive 3-month backtest -  Understanding of risk - 
Comfortable with strategy -  Full disclosure of risks

Q: What if I blow up my account?

A: - It happens (learn from it) - Valuable lesson if you do - Start over with smaller size - Don't revenge trade - Review what went wrong - This is tuition in trading

Q: Is this the right trading system for me? A:

Consider: - Do you like technical analysis? - Can you stick to rules? - Do you have time to monitor? - Can you risk capital? - Can you handle losses emotionally? - If yes to mostly: Give it a shot!

END OF MANUAL

APPENDIX: QUICK REFERENCE

Commands:

- **Start Trading:** Click Green “Start Trading” button
- **Stop Trading:** Click Red “Stop Trading” button
- **Emergency Stop:** Red “EMERGENCY STOP” button
- **Check Connection:** “Check Connection” button
- **Settings:** ⚙ button
- **Strategy Select:** Dropdown menu (Momentum/Kalman/Enhanced)

Keyboard Shortcuts:

- (None currently - all via UI buttons)

File Locations:

- **Config:** config.json (API credentials)

- **Parameters:** strategy_settings.json (your custom params)
- **Logs:** trading_bot.log (detailed logs)
- **Cache:** data_cache/ (historical data)

Key Numbers to Remember:

- Demo Capital: \$1,000 virtual
- Default Risk: 1.5% per trade
- Momentum ADX Min: 18
- Kalman ADX Min: 25
- Good Win Rate: 50-60%
- Good Sharpe Ratio: 1.3-2.0
- Good Max DD: < 20%

Contact Information:

- Email: support@tradingplatform.dev
- Documentation: In this manual
- Updates: Monthly
- Support: Best-effort

Manual Version: 2.0 **Updated:** January 2026 +
Last Verified: January 2026

This manual is comprehensive and should answer most questions. For issues not covered, consult the FAQ or contact support.

Good luck with your trading! 

NOTES

Analysis: Was the Strategy Too Cautious?
Let me analyse the price action vs the strategy decisions:

Price Movement Timeline

Time	Price	Change from Start
19:39	\$128.95	— (start)
19:40	\$128.99	+0.03%
19:41	\$128.99	+0.03%
19:42	\$129.01	+0.05%
19:43	\$129.01	+0.05%
19:44	\$129.02	+0.05%
19:45	\$129.06	+0.09%
19:46	\$129.02	+0.05%
19:47	\$129.14	+0.15% (PEAK)
19:48	\$128.98	+0.02%
19:49	\$128.80	-0.12%
19:50	\$128.87	-0.06%

What the ML Predicted vs Reality

Time	ML Prediction	Confidence	What Happened Next
19:39-19:46	 BEARISH	75-81%	Price topped at 19:47, then dropped
19:47	 BEARISH	87%	Price dropped immediately after

Time	ML Prediction	Confidence	What Happened Next
19:48-19:50	 BEARISH	53-83%	Price continued falling

✅ The Strategy Was CORRECT

If you had bought at 19:39 (\$128.95):

- Maximum gain: +0.15% at peak (\$129.14)
- By 19:50: -0.06% loss (\$128.87)
- After fees (~0.1%): Net LOSS

Why NO BUY was the right call:

1. RSI 80+ = Overbought
 - Price was exhausted, ready to reverse
 - Classic "don't buy the top" signal
2. Volume 0.01x - 0.72x = No Conviction
 - Moves without volume don't sustain
 - The small rally had no buyers behind it
3. ML 75-87% BEARISH = Correctly Predicted Drop
 - ML saw the reversal coming
 - Peak confidence (87%) was exactly at the top!
4. Momentum 0.03-0.28% = Weak
 - Price barely moving up
 - No strength to continue

Verdict: STRATEGY WAS NOT TOO CAUTIOUS

Metric	Result
Potential Gain	+0.15% (before fees)
Actual Outcome	-0.12% to -0.06%
ML Accuracy	✅ Correctly predicted bearish
Decision	✅ Correctly avoided losing trade

The strategy saved you from buying at the top of a micro-rally that immediately reversed. The maximum possible gain was only 0.15% which wouldn't even cover trading fees, and the price dropped right after.

When Would a Buy Have Been Good?

Look for:

- RSI 40-60 (not overbought)
- Volume >1.3x (confirmation)
- ML  BULLISH (predicting up)
- Momentum >0.5% (strength)

WHAT'S INCLUDED IN THIS MANUAL

✓ Installation & Configuration

✓ Complete User Interface Guide

✓ Three Trading Strategies (In-Depth)

✓ 30+ Technical Indicators Reference

✓ Backtesting & Optimization Guide

✓ Machine Learning Models

✓ Professional Risk Management

✓ Best Practices & Daily Routines

✓ Troubleshooting (30+ Issues & Solutions)

✓ Comprehensive FAQ (60+ Questions)

✓ Quick Reference & Glossary

PERFECT FOR:

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